

# TrES-2b

R-1108

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-2\_140526 · created 2026-07-30T20:29:14+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456803.78259 +/- 0.0003 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1116 +/- 0.0096                |
| Transit depth (Rp/Rs)^2                           | 1.25 +/- 0.21 [%]                |
| Orbital Inclination (inc)                         | 85.39 +/- 0.64                   |
| Airmass coefficient 1 (a1)                        | 1.0013 +/- 0.0097                |
| Airmass coefficient 2 (a2)                        | 0.0002 +/- 0.0042                |
| Scatter in the residuals of the lightcurve fit is | 0.97 %                           |
| Best Comparison Star                              | #3 - [294, 380]                  |
| Optimal Aperture                                  | 3.34                             |
| Optimal Annulus                                   | 13.29                            |
| Transit Duration (day)                            | 0.0906 +/- 0.0064                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                          |
|--------------------------------------|----------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1116 $\pm$ 0.0096 vs 0.1254 (deviation 1.22 $\sigma$ ) |
| Duration fit vs published            | 0.0906 d vs 0.0746 d (deviation 2.12 $\sigma$ )          |
| Mid-time O-C                         | 0.7 min                                                  |
| Depth SNR                            | 9.9                                                      |
| Residual scatter                     | 0.97 %                                                   |

### Light curve

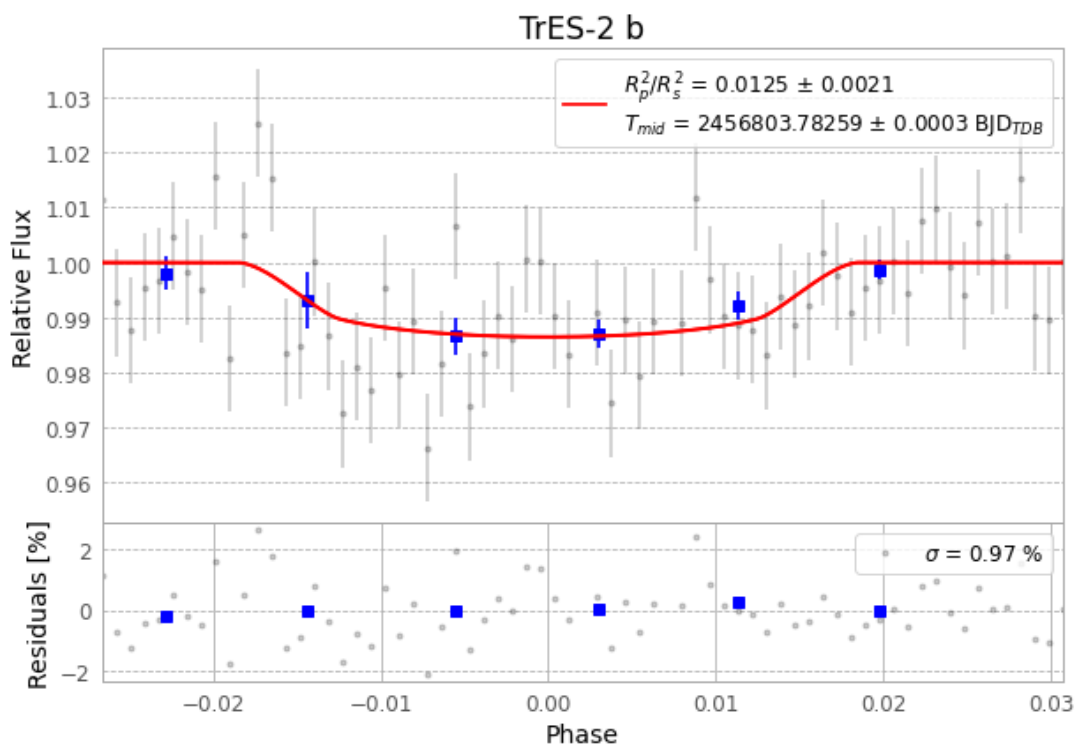


Figure 1 — FinalLightCurve\_TrES-2 b\_2014-05-25.png (SHA-256 ea6f3baf71b144b1...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [234, 222], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 2ed907fb632bcba6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

69 FITS frames (45 MB), TRES-2140526050912.FITS → TRES-2140526083313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140526.log (d8d46415e611...), FinalLightCurve\_TrES-2 b\_2014-05-25.png (ea6f3baf71b1...), FinalLightCurve\_TrES-2 b\_2014-05-25.csv (bcd4234e6ae7...)

## Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:29Z.